R2.3 How far? The extent of chemical change

Equilibrium and Gibbs energy (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to calculate the value of the Gibbs energy change ΔG from the equilibrium constant for a reaction.

If a rock is rolled down a hill, where does it stop? The rock will stop rolling when it gets to the lowest point which is the lowest energy position as shown in **Figure 1**. To continue rolling up the other side would require an input of energy as would returning to its original position. A rock rolling down a hill is comparable to a spontaneous reaction section R1.4.2–3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-spontaneous-changes-hl-id-44766/) which is a reaction that proceeds without the addition of energy once the initial energy barrier has been overcome.

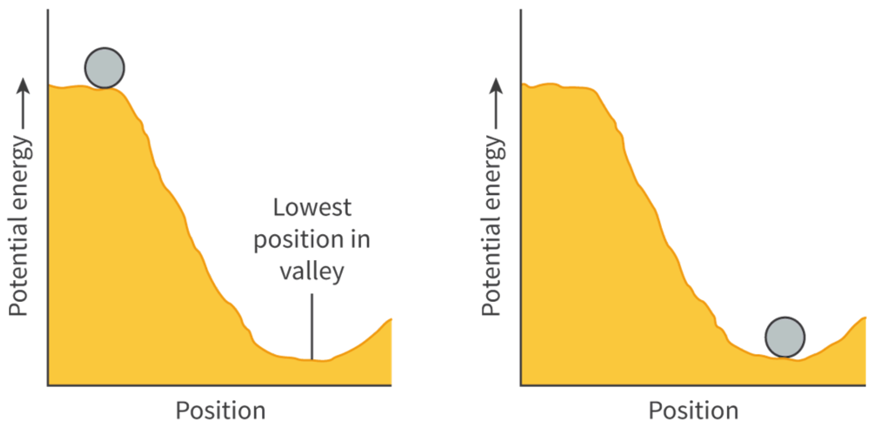

Figure 1. A rock rolling down a hill reaches the lowest energy point. This is analogous to a chemical reaction proceeding towards equilibrium.

More information for figure 1

The image consists of two graphical diagrams side by side, both illustrating potential energy in relation to position. In the left diagram, a rock is positioned at the top of a hill with high potential energy and begins to roll down a slope. The slope dips into a valley labeled as the "Lowest position in valley," where the potential energy is at its minimum. The right diagram shows the rock at rest in this valley, reaching equilibrium at the lowest energy point, illustrating a stable position with low potential energy. Both diagrams use an upward triangular shape to indicate the change in potential energy and demonstrate a physical analogy to a chemical reaction reaching a state of equilibrium.

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Figure 2 and **Figure 3** shows the change in Gibbs energy (see section R1.4.2–3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-spontaneous-changes-hl-id-44766/)) for a spontaneous and a non-spontaneous reaction, respectively. For both reactions the Gibbs energy reaches a minimum at which point the reaction is at equilibrium and the Gibbs energy change, ΔG, is zero. However, one important difference is the point in the reaction where the minimum position of Gibbs energy lies.

Study skills

A reaction at equilibrium has a minimum value of Gibbs energy and a maximum value of entropy.

For a spontaneous reaction (**Figure 2**), the minimum value of Gibbs energy lies further towards the products side meaning that the equilibrium mixture will contain mostly products. The equilibrium constant for a spontaneous reaction is greater than one, with its value increasing as a reaction becomes more spontaneous.

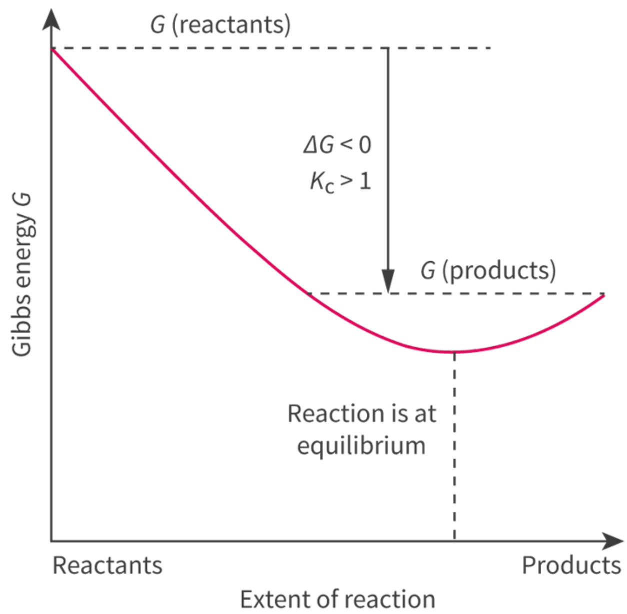

Figure 2. For a spontaneous reaction, the minimum point of Gibbs energy lies towards the products side.

More information for figure 2

The image is a graph depicting the Gibbs energy (G) as a function of the extent of reaction. The X-axis is labeled 'Extent of reaction' and is divided into 'Reactants' on the left and 'Products' on the right. The Y-axis is labeled 'Gibbs energy G'. The graph shows a curve that starts high at the reactants side, indicating an initial high Gibbs energy. It slopes downward and reaches a minimum point towards the products side, indicating a lower Gibbs energy there. This signifies that the reaction is spontaneous. The energy difference is marked as ΔG < 0. Text in the graph indicates "Kc > 1" and "Reaction is at equilibrium" at the lowest point of the curve, which is closer to the products side than the reactants.

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For a non-spontaneous reaction (**Figure 3**), the minimum value of the Gibbs energy lies further towards the reactants side meaning that the equilibrium mixture will contain mostly reactants. The equilibrium constant for a non-spontaneous reaction is less than one, with its value decreasing as a reaction becomes less spontaneous.

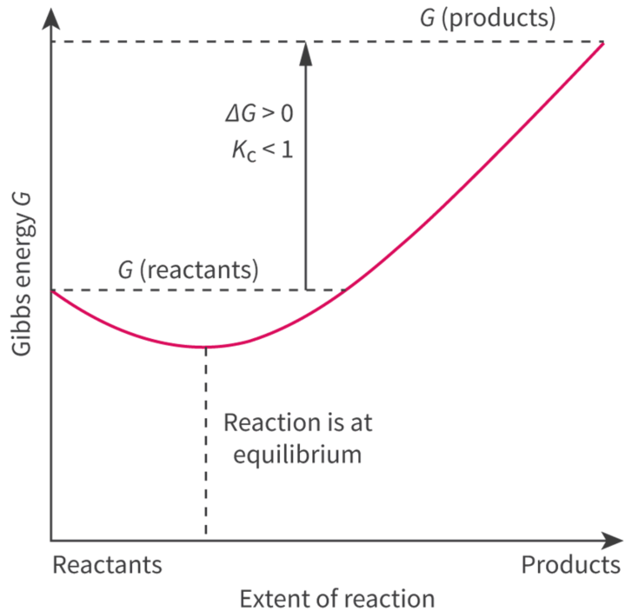

Figure 3. For a non-spontaneous reaction, the minimum point of Gibbs energy lies towards the reactants side.

More information for figure 3

The graph represents the relationship between Gibbs energy (G) and the extent of a reaction for a non-spontaneous process. The X-axis is labeled as "Extent of reaction" and ranges from "Reactants" to "Products." The Y-axis is labeled as "Gibbs energy G."

The graph features a curve that starts from the left at a higher energy level, decreases to a minimum point, and then rises again towards the right. The minimum point of the Gibbs energy curve lies towards the reactants side, indicating the position of equilibrium.

Annotations on the graph include a vertical dashed line at the minimum point labeled "Reaction is at equilibrium," showing where the equilibrium position is. The labels "G (reactants)" and "G (products)" are positioned horizontally on the left and right sides, respectively.

Additional annotations include ΔG > 0 and Kc < 1, indicating that the reaction is non-spontaneous and the equilibrium constant (Kc) is less than one, consistent with a reaction where the products are not favored.

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The relationship between the ΔG, the equilibrium constant K_c and the composition of the equilibrium mixture is summarised in **Table 1**.

Table 1. The relationship between the ΔG, the equilibrium constant K_c and the compositio equilibrium mixture.

$\Delta G^{\ominus}$	K_c	Composition of reaction mixture
< 0	> 1	Higher concentration of products
> 0	< 1	Higher concentration of reactants
0	1	Neither reactants nor products favoured

Making connections

How can Gibbs energy be used to explain whether the forward or backward reaction is favoured before reaching equilibrium (see subtopic R1.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-hl-id-43477/))?

Calculating ΔG from the equilibrium constant

Recall that the Gibbs energy change, ΔG, can be used to determine whether a reaction will occur spontaneously at a particular temperature (see section R1.4.2–3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-spontaneous-changes-hl-id-44766/)). In this section you will learn about the relationship between the Gibbs energy change, ΔG, and the equilibrium position. The Gibbs energy change can be calculated from the value of the equilibrium constant, K_c, for a reaction using the following equation, which can be found in section 1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/) of the DP chemistry data booklet.

Table 2. Equation for standard Gibbs energy change.

Equation	Symbols	Units
$\Delta G^{\ominus} = -RT \ln K$	$\Delta G^{\ominus}$ is the standard Gibbs energy change	joules (J)
	R is the universal gas constant ($8.31\ J\ K^{-1}\ mol^{-1}$)	joules per kelv mole ($J\ K^{-1}\ mol$
	T is the temperature	kelvin (K
	$\ln K$ is the natural log of the equilibrium constant, K_c	no unit

In this equation $\Delta G^{\ominus}$ has units of $J\ mol^{-1}$, to convert to $kJ\ mol^{-1}$, divide by 1000.

The equation can be rearranged to calculate the value of the equilibrium constant, K_c, if the Gibbs energy change is known:

$$K = e^{-\frac{\Delta G}{RT}}$$

Next, we will look at some example calculations, starting with calculating the Gibbs energy change, Δ*G*, for a reaction using the value of the equilibrium constant.

Nitrogen and oxygen react together to form nitrogen monoxide as shown.

$$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$$

The value of the equilibrium constant for this reaction at 150 °C is 1.60×10^{-3}.

Determine the value of the Gibbs energy change, Δ*G*, in kJ mol^{-1}.

$$\Delta G^{\ominus} = -RT \ln K$$

Substitute the values into the equation, making sure that the temperature is in kelvin (K).

$$\begin{aligned} \Delta G &= -8.31 \times (150.0 + 273.15) \times (\ln 1.60 \times 10^{-3}) \\ &= 22\,630 \text{ J mol}^{-1} \end{aligned}$$

Convert to kJ mol^{-1} by dividing by 1000:

$\Delta G = 22.6$ kJ mol^{-1} (to three significant figures)

The positive Δ*G* value means the reaction is non-spontaneous and the equilibrium position lies to the left which is shown by the small value of the equilibrium constant.

Worked example 1

Calculate the value of the Gibbs energy change, Δ*G*, in kJ mol^{-1}, for this reaction at 298 K.

$$N_2O_4(g) \rightleftharpoons 2NO_2(g) \qquad K = 0.140$$

$$\Delta G^{\ominus} = -RT \ln K$$

$$\begin{aligned} \Delta G^{\ominus} &= -8.31 \times 298 \times \ln 0.140 \\ &= 4.87 \times 10^3 \text{ J mol}^{-1} \\ &= 4.87 \text{ kJ mol}^{-1} \end{aligned}$$

Next we will look at how to calculate the value of the equilibrium constant for a reaction using the value of the Gibbs energy change, Δ*G*.

The Haber process has a Gibbs energy change, Δ*G*, of –33.0 kJ mol^{-1} at 25.0 °C.

Calculate the value of the equilibrium constant at this temperature.

$$K = e^{-\frac{\Delta G}{RT}}$$

Substitute in the values and calculate the value of *K*. Note the value of the Δ*G* is converted to J mol^{-1} by multiplying by 1000 and the temperature is converted to kelvin by adding 273.15.

$$\begin{aligned} K &= e^{-\frac{-33.0 \times 1000}{8.31 \times (25.0 + 273.15)}} \\ &= e^{-(-13.3)} \\ &= 5.97 \times 10^5 \end{aligned}$$

This value of K is expected because the ΔG is negative meaning that the equilibrium position lies to the right and the equilibrium mixture contains a higher concentration of products.

Worked example 2

Calculate the value of the equilibrium constant for this reaction at 298 K.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \qquad \Delta G = -142 \text{ kJ mol}^{-1}$$

$$K = \mathrm{e}^{-\frac{\Delta G}{RT}}$$

Don't forget to convert the ΔG to J mol^{-1}. The temperature is already given in Kelvin.

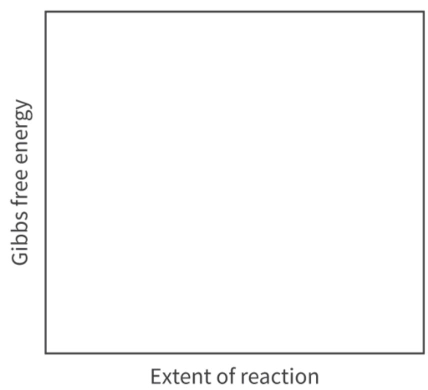
$$\begin{aligned} K &= \mathrm{e}^{-\frac{-142\times1000}{8.31\times298}} \\ &= \mathrm{e}^{-(-57.3)} \\ &= 7.67 \times 10^{24} \end{aligned}$$

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

1. On the axis provided below, sketch the change in Gibbs energy for a spontaneous and a non-spontaneous reaction. On each graph, label the equilibrium position.

   Spontaneous reaction



More information

The image contains a graph with the X-axis labeled "Extent of reaction" and the Y-axis labeled "Gibbs free energy". This graph is intended to show the change in Gibbs energy for both a spontaneous and a non-spontaneous reaction.

The task is to sketch:

a. A spontaneous reaction: Typically, the curve would show a decrease in Gibbs free energy as the extent of the reaction progresses, indicating that the reaction releases energy and moves towards equilibrium.
b. A non-spontaneous reaction: The curve would likely show an increase or no significant change in Gibbs free energy, suggesting that the reaction requires added energy to proceed and does not naturally progress towards equilibrium.

The equilibrium position on each graph should be clearly labeled, marking the point where the reaction is balanced with no net change in Gibbs free energy.

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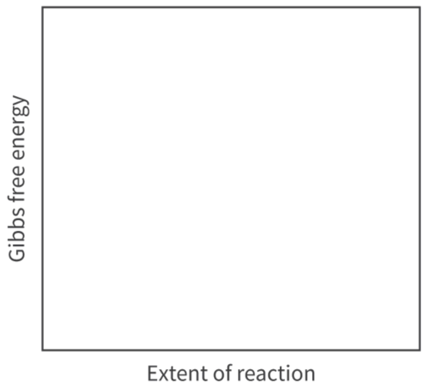
Non-spontaneous reaction



More information

The image is a graph plotting 'Gibbs free energy' (y-axis) against 'extent of reaction' (x-axis). The graph represents how the Gibbs free energy changes as a reaction progresses. The axes are labeled with pertinent scientific terms essential for interpreting chemical thermodynamics. However, no specific data points, curves, or numerical values are visibly marked on the graph.

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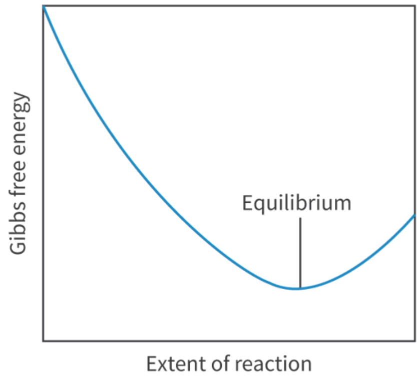
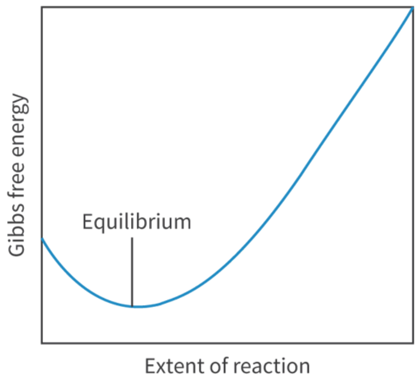
2. Calculate the Gibbs energy change, ΔG, in kJ mol^{-1}, for a reaction that has an equilibrium constant of 54.3 at 430 °C.
3. Calculate the equilibrium constant for a reaction that has a Gibbs energy change, ΔG, of 41.8 kJ mol^{-1} at 25.0 °C.

1. Spontaneous reaction



Non-spontaneous reaction



2. $\Delta G = -RT \ln K$

$$\begin{aligned}\Delta G &= -8.31 \ \times (431 + 273.15) \times (\ln 54.3) \\ &= -23369 \text{ J mol}^{-1}\end{aligned}$$

Convert to kJ mol^{-1} by dividing by 1000:

$$\Delta G = -23.4 \text{ kJ mol}^{-1}$$

3.

$$K_c = \mathrm{e}^{-\frac{\Delta G}{RT}}$$

$$\begin{aligned}K_c &= \mathrm{e}^{-\frac{11.8 \times 1000}{8.31 \times (25.0 + 273.15)}} \\ &= \mathrm{e}^{-(16.9)} \\ &= 4.58 \times 10^{-8}\end{aligned}$$